MAP & Conjugate Priors

MLE, with a prior’s vote

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Mathematical Foundations for AI · IIT Gandhinagar

Three heads in three flips

Where Lecture 14 left us

L14 fixed the data, varied the parameter, and maximized the likelihood.

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} P(D \mid \theta)$$

- The coin: n_H heads in n flips $\Rightarrow \hat{\theta}_{\text{MLE}} = n_H/n$ — count and divide.
- The keystone: least squares *is* Gaussian MLE — **MSE = Gaussian NLL**.
- The insight line: *every loss function is a negative log-likelihood in disguise.*

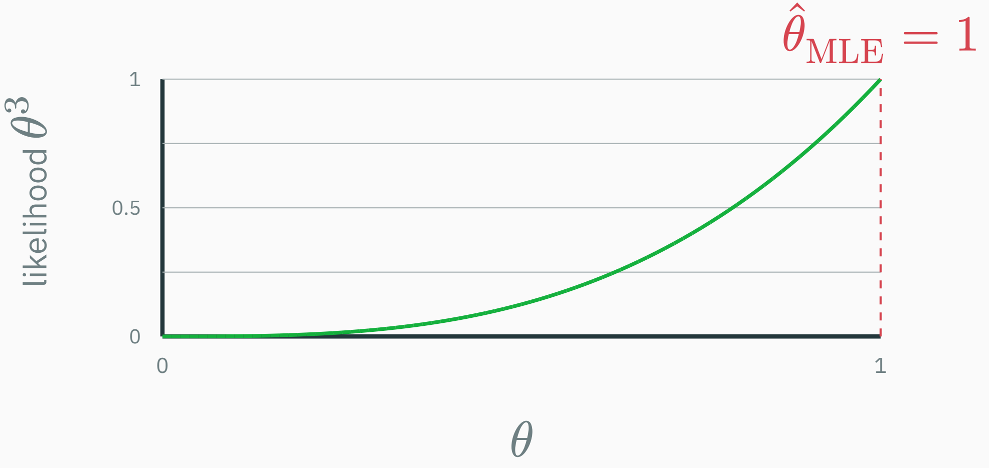
Today we break this beautiful machine — with a dataset of size **three**.

The cliffhanger: three flips, three heads

V VISUAL

You flip a coin from your pocket three times. Heads, heads, heads. MLE does what it always does:

$$\hat{\theta}_{\text{MLE}} = \frac{n_H}{n} = \frac{3}{3} = 1.0$$



The estimate is not “heads-ish”. It is $\theta = 1$: **this coin can never land tails.**

Would you take the bet?

$\hat{\theta} = 1$ is a checkable claim: it prices "at least one tail in the next 100 flips" at **exactly zero**.

- Your gut refuses the bet. Pocket coins are almost always near-fair.
- The three flips are real evidence — but they are **three flips**.
- And your gut isn't cheating: it carries **years of previous coins**. That knowledge is data too — it just arrived before today.

MLE has no slot for what you knew before the data. Today we build that slot: the prior.

The same bug ships in real AI

System	The three-heads bug, at work
spam filter	“invoice” never appeared in the spam folder → $P(\text{invoice} \mid \text{spam}) = 0$ → one word acquits any email, forever
drug trial	3 successes in 3 patients → “this drug *cannot* fail”
language model	a letter pair never seen in training → probability 0 → $\log 0 = -\infty$ loss on any text containing it

Plant for L26: we will build exactly that character language model — and rescue it with *Dirichlet smoothing*, which is today’s mathematics wearing a bigger alphabet.

What would a fix even look like?

Three demands, straight from your intuition:

1. Tiny datasets must **not** produce absolute certainty.
2. Enough data must be able to **overrule** any hunch.
3. The hunch itself must be **adjustable** — a fairness zealot and a con-artist-spotter should be able to encode different starting beliefs.

Today's route: **Bayes' rule** (the legal merge of belief and evidence) → **MAP** (MLE with the prior voting) → **conjugacy** (the trick that makes the update one line).

Learning outcomes

By the end of this lecture you will be able to:

1. Read Bayes' rule as a **belief update**, naming all four factors and their jobs.
2. Compute **MAP** estimates and say exactly when MAP $=$ MLE.
3. Use the **Beta distribution** as a prior over probabilities, reading α, β as pseudo-counts.
4. Derive ⭐ the **Beta–Bernoulli conjugate update**, Beta$(\alpha + n_H, \beta + n_T)$.
5. Update **sequentially** as data streams in — and explain why order can't matter.
6. State the **Gaussian–Gaussian** posterior as a precision-weighted average.
7. Unmask **L2 regularization** as a Gaussian prior (and L1 as a Laplace prior).

Bayes’ rule: the belief-update machine

The four factors, named

One rule merges what you believed with what you saw:

$$P(\theta \mid D) = \frac{P(D \mid \theta) \cdot P(\theta)}{P(D)}$$

Factor	Name	Its job
$P(\theta \mid D)$	**posterior**	belief about θ *after* the data — the answer
$P(D \mid \theta)$	**likelihood**	how loudly the data argues for each θ — L14's object
$P(\theta)$	**prior**	belief about θ *before* the data — new today
$P(D)$	**evidence**	one number that makes the left side a distribution

Numbers first: a bag of coins

A bag holds 10 coins: **9 fair** ($\theta = 0.5$), **1 rigged** ($\theta = 0.9$). You draw one at random, flip it 3 times: **3 heads**. Did you draw the rigged one?

Two hypotheses → Bayes' rule is a two-row table:

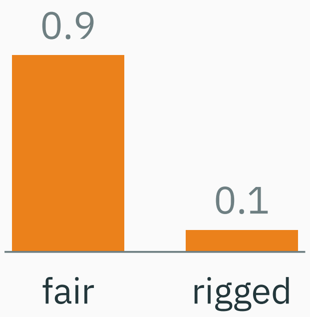
Coin	prior	likelihood θ^3	prior $\times$ lik	posterior
fair	0.9	$0.5^3 = 0.125$	0.1125	$0.1125/0.1854 = \mathbf{0.607}$
rigged	0.1	$0.9^3 = 0.729$	0.0729	$0.0729/0.1854 = \mathbf{0.393}$

Three heads moved "rigged" from **10% to 39%** — suspicion, not a conviction. No factor of the update is mysterious: multiply, then rescale the two numbers to sum to 1.

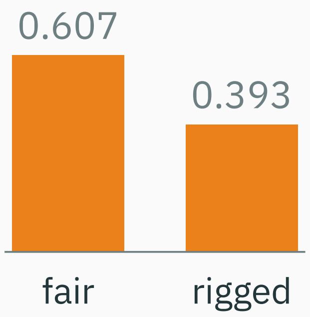
The update, drawn — and coded

V VISUAL

before: prior



after 3 heads: posterior




```
prior = np.array([0.9, 0.1])          # P(fair), P(rigged)
like  = np.array([0.5**3, 0.9**3])    # P(HHH | each coin)
post  = prior * like                  # multiply...
post /= post.sum()                    # ...normalize -> [0.607, 0.393]
```

Every Bayesian computation you will ever run is these four lines, scaled up.

The evidence is bookkeeping

The denominator $P(D) = \sum_h P(D \mid h)\, P(h)$ adds over **hypotheses** — no θ survives inside it. It rescales the curve; it cannot move the peak.

$P(\theta \mid D) \propto P(D \mid \theta) \cdot P(\theta)$

Posterior $\propto$ likelihood $\times$ prior. The evidence is the normalization constant that makes the posterior integrate to one.

That $\propto$ sign is about to do a lot of honest work — hold on to it.

MAP: let the prior vote

Same recipe as MLE — but climb the **posterior** instead of the likelihood:

$$\theta_{\text{MLE}} = \arg\max_{\theta} P(D \mid \theta) \qquad \theta_{\text{MAP}} = \arg\max_{\theta} P(D \mid \theta) \cdot P(\theta)$$

In log-space (products → sums — L2's rule, as always):

$$\theta_{\text{MAP}} = \arg\max_{\theta} \left[\underbrace{\log P(D \mid \theta)}_{\text{data's vote}} + \underbrace{\log P(\theta)}_{\text{prior's vote}} \right]$$

MAP = maximum a posteriori. The prior enters as one additive vote on the log scale — it bends the landscape before the climb starts.

A flat prior changes nothing

Take $P(\theta) = 1$ on $[0, 1]$ — every bias equally believable. Then $\log P(\theta) = 0$:

$$\theta_{\text{MAP}} = \arg\max_{\theta} [\log P(D \mid \theta) + 0] = \theta_{\text{MLE}}$$

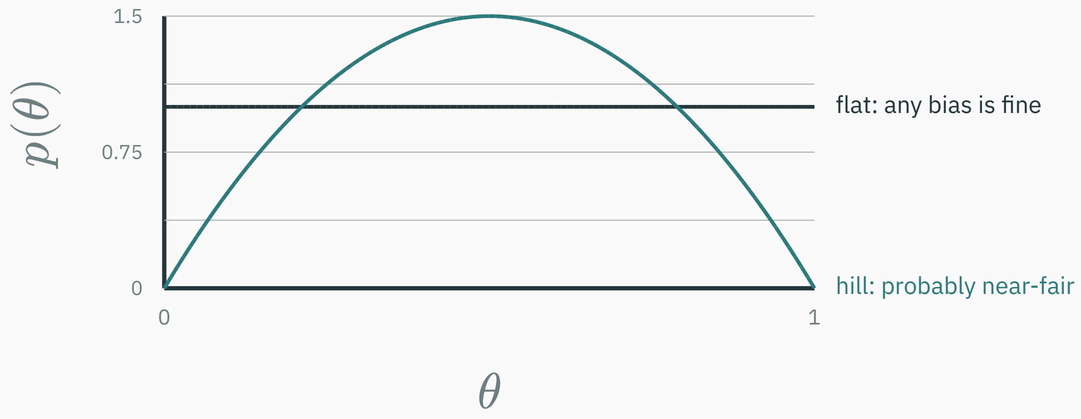
MLE is MAP with a flat prior. MLE was never "assumption-free" — its assumption was *flatness*.

So the real question is never "prior or no prior?" — it is **which** prior. It matters most when data is scarce, skewed, or expensive; it stops mattering when data is plentiful (proof by formula, later today).

The coin gets a prior

A prior must live on $[0, 1]$

θ is a probability, so a prior is a **density on $[0, 1]$** (L12: continuous belief is area). Two candidates:



The hill is $p(\theta) \propto \theta(1-\theta)$ — zero at the impossible extremes, gentle peak at $1/2$. (It has a family name; we'll meet the family in a few slides.)

Worked: the flat prior

Multiply the likelihood of three heads by the flat prior:

$$P(\theta \mid D) \propto \underbrace{\theta^3}_{\text{likelihood}} \cdot \underbrace{1}_{\text{prior}} = \theta^3$$

Climbing θ^3 on $[0, 1]$ ends at the right wall:

$$\theta_{\text{MAP}} = 1.0 = \theta_{\text{MLE}}$$

The absurd bet survives a flat prior — flatness has no opinion to add. To fix the pathology we must actually **hold** a belief.

Worked: the hill prior, by hand

Same three heads, hill prior $p(\theta) \propto \theta(1-\theta)$:

$$P(\theta \mid D) \propto \theta^3 \cdot \theta(1-\theta) = \theta^4(1-\theta)$$

Maximize with one derivative (drop constants; they can't move a peak):

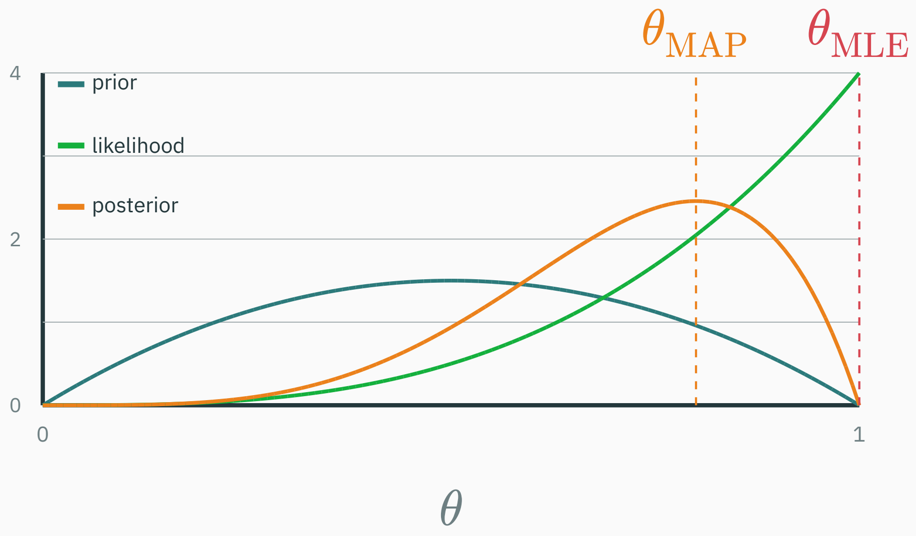
$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left[\theta^4(1-\theta)\right] = 4\theta^3 - 5\theta^4 = \theta^3(4-5\theta) = 0$$

$$\Rightarrow \quad \theta_{\mathrm{MAP}} = \frac{4}{5} = 0.8$$

Same three heads: the flat prior says 1.00; a gentle near-fair belief says 0.80.

Prior $\times$ likelihood $=$ posterior, drawn

V VISUAL



Likelihood rescaled to area 1 so the three shapes share one axis.

The posterior sits **between** the prior's peak (0.5) and the likelihood's peak (1.0) — a negotiated settlement, weighted by how insistent each side is.

Check it by computer

No calculus, just a grid — the same argmax, numerically:

```
th   = np.linspace(0, 1, 1001)
flat = th**3                          # likelihood x flat prior
hill = th**3 * th * (1 - th)          # likelihood x hill prior

th[flat.argmax()], th[hill.argmax()]     # -> (1.0, 0.8)
```

- Shapes only — no normalizing constants anywhere, because `argmax` ignores scale.
- This grid trick works for **any** prior you can evaluate — calculus optional. (T7 uses it to cross-check every derivation.)

Checkpoint: MLE vs MAP

Q QUESTION

A coin shows 9 heads in 10 flips. With a flat prior on θ, the MAP estimate is...

A. 0.50 — the prior always pulls to fair

B. 0.90 — exactly the MLE

C. 0.83 — pulled part-way toward $1/2$

D. 1.00 — MAP always sharpens the MLE

Answer: MLE vs MAP

A ANSWER

Correct: B — $\theta_{\text{MAP}} = 0.90 = \theta_{\text{MLE}}$

Why: A flat prior adds $\log 1 = 0$ to the log-posterior — the argmax cannot move, so MAP $=$ MLE $= 9/10$. Answer C (≈ 0.83) is what the *hill* prior Beta$(2, 2)$ would give — right instinct, wrong prior; A forgets the data entirely; D isn't a thing MAP does.

The Beta family as a prior over probabilities

The hill has a family name

Tilt the exponents and the hill becomes a whole family of beliefs on $[0, 1]$:

$$\mathrm{Beta}(\theta; \alpha, \beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \qquad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

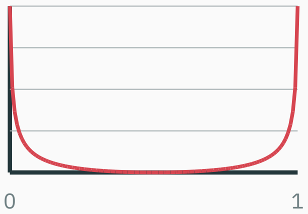
- $\Gamma(k) = (k-1)!$ for positive integers, and the gamma function extends this definition so α and β need not be integers.
- Our hill $\propto \theta(1-\theta)$ is $\mathrm{Beta}(2, 2)$; the flat prior is $\mathrm{Beta}(1, 1)$. (L12′s zoo listed Beta as “the star of L15” — here it is.)

$B(\alpha, \beta)$ exists so the area is 1. We will **never** integrate it in this lecture — inspection will hand it to us for free.

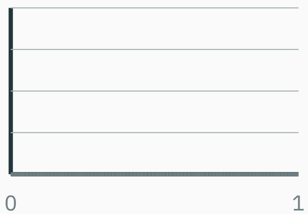
The shape gallery

All dials from $\{0.5, 1, 2, 5\}$ — one family, five personalities:

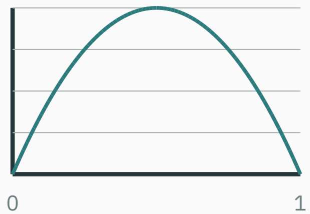
Beta(0.5, 0.5) — extremist



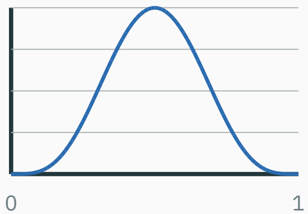
Beta(1, 1) — agnostic



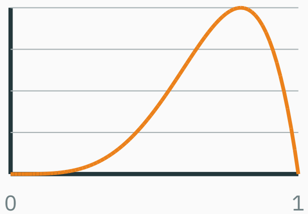
Beta(2, 2) — gentle hunch



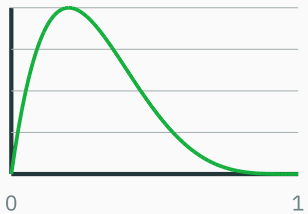
Beta(5, 5) — firmer hunch



Beta(5, 2) — leans heads



Beta(2, 5) — leans tails



One knob pair covers superstition, indifference, hunches, conviction, and lopsided suspicion.

Read α, β as pseudo-counts

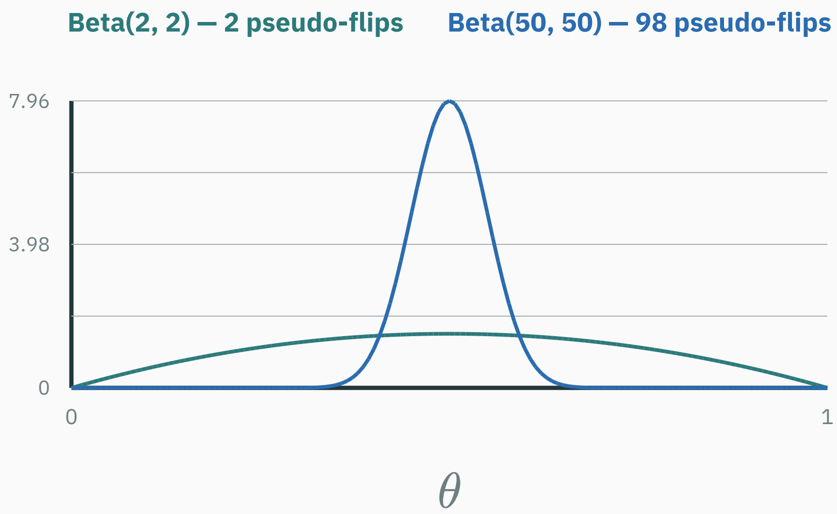
For $\alpha, \beta > 1$ the peak sits at $\text{mode} = (\alpha - 1)/(\alpha + \beta - 2)$ — exactly the MLE of a record with $\alpha - 1$ heads and $\beta - 1$ tails.

Prior	Imaginary record	Peak
Beta$(1, 1)$	0 H, 0 T — no memory at all	flat
Beta$(2, 2)$	1 H, 1 T — the gentlest fairness hint	0.5
Beta$(5, 2)$	4 H, 1 T — suspects a heads bias	0.8
Beta$(50, 50)$	49 H, 49 T — fairness conviction	0.5

A prior is a memory of flips you never made — $\alpha - 1$ imaginary heads, $\beta - 1$ imaginary tails.

How sure? $\alpha + \beta$ is the stake

Same center, different bankroll: mean $= \alpha/(\alpha+\beta)$ fixes *where*; the total $\alpha+\beta$ fixes *how firmly*.



Real flips will have to **out-vote the imaginary ones** — a Beta$(50, 50)$ believer needs far more evidence to move. That's the whole coming fight, stated in advance.

Play the update before we derive it

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/bayesian-posterior

Drag the prior's α, β, stream in evidence across 8 experiments, and watch prior $\times$ likelihood $=$ posterior re-render live.

INTERACTIVE

↗ seeing-theory.brown.edu/bayesian-inference/index.html

Seeing Theory · Bayesian Inference — flip a coin one toss at a time and watch a Beta posterior fill in.

Both toys are running one identity — the one we now derive by hand.

Conjugacy: the one-line update

⭐

Set up the multiplication

D DERIVATION

Data: n_H heads, n_T tails ($n = n_H + n_T$). L14's likelihood, and a Beta(α, β) prior — constants dropped, shapes kept ($\propto$ absorbs $1/B$):

$$P(\theta \mid D) \propto \underbrace{\theta^{n_H}(1-\theta)^{n_T}}_{\text{likelihood: real flips}} \cdot \underbrace{\theta^{\alpha-1}(1-\theta)^{\beta-1}}_{\text{prior: imaginary flips}}$$

Both factors speak the **same language**: powers of θ and powers of $(1-\theta)$. That coincidence is about to do all the work.

Exponents simply add

D DERIVATION

Same base, so multiplication is addition upstairs:

$$P(\theta \mid D) \propto \theta^{n_H+\alpha-1}\,(1-\theta)^{n_T+\beta-1}$$

Stare at it: this is **the Beta shape again**, with new dials

$$\alpha' = \alpha + n_H, \qquad \beta' = \beta + n_T$$

Real heads landed on the imaginary-heads pile; real tails on the imaginary-tails pile. The data didn't change the *kind* of belief — only its *counts*.

Normalize by inspection

D DERIVATION

We hold a function $\propto \theta^{\alpha'-1}(1-\theta)^{\beta'-1}$ that must integrate to 1 (it's a posterior). Among all rescalings $c \cdot \theta^{\alpha'-1}(1-\theta)^{\beta'-1}$, **exactly one** has area 1 — and $\mathrm{Beta}(\alpha', \beta')$ is that one, by definition.

So the constant is forced, no integral computed:

$$\boxed{\mathrm{Beta}(\alpha,\beta)\ \text{prior} + n_H\ \text{heads}, n_T\ \text{tails} \quad \Rightarrow \quad \text{posterior} = \mathrm{Beta}(\alpha+n_H, \beta+n_T)}$$

Honesty note: "by inspection" means we recognized a member of a family whose normalizer Euler already paid for. The integral happened once, centuries ago — never again at update time.

Conjugacy, named

Definition. A prior family is **conjugate** to a likelihood if the posterior always lands back in the same family.

- Beta in → Bernoulli data → Beta out: the Beta family is conjugate to coin flips.
- The entire update is **two integer additions**: $\alpha \leftarrow \alpha + n_H$, $\beta \leftarrow \beta + n_T$. No integral, no grid, no optimizer.
- Our worked example, re-told: Beta$(2, 2)$ + 3 heads → Beta$(5, 2)$ — a pseudo-record of 4 H, 1 T, peak at $4/5$. The posterior **is** a merged memory of real and imaginary flips.

The MAP formula drops out

The posterior is Beta$(\alpha + n_H, \beta + n_T)$, and we know where a Beta peaks:

$$\theta_{\text{MAP}} = \frac{n_H + \alpha - 1}{n + \alpha + \beta - 2}$$

- Check: 3 heads, Beta$(2, 2)$: $(3+1)/(3+2) = 4/5 = 0.8$ — matches the hand derivative. ✓
- Check: flat prior $\alpha = \beta = 1$: $\theta_{\text{MAP}} = n_H/n = \theta_{\text{MLE}}$. ✓

Read it aloud: real heads + imaginary heads, over real flips + imaginary flips.

Stress test: 9 heads, 1 tail

Ten flips, nine heads ($\theta_{\text{MLE}} = 0.9$). Three believers meet the same data:

Prior	θ_{MAP}	Reading
Beta$(1, 1)$ — agnostic	$9/10 = 0.900$	no opinion → data speaks alone
Beta$(2, 2)$ — gentle hunch	$10/12 \approx 0.833$	two pseudo-flips nudge back toward fair
Beta$(50, 50)$ — conviction	$58/108 \approx 0.537$	98 pseudo-flips: ten real ones barely dent it

Same data, three defensible answers — the prior is a **published, criticizable assumption**, not a hidden thumb on the scale. (Which believer is being unreasonable? That argument is now about α, β — in the open.)

The conjugate catalog

The same one-line-update miracle exists beyond coins:

Likelihood	Conjugate prior	Posterior update
Bernoulli (coin)	Beta	$\alpha \leftarrow \alpha + n_H$, $\beta \leftarrow \beta + n_T$
Categorical (d faces)	Dirichlet	add each face's count to its pseudo-count
Gaussian mean (σ known)	Gaussian	precision-weighted average — next section

Continuity ledger — planted today, paid off in L26: the Dirichlet row is Beta's d-sided sibling. When L26′s character language model meets an unseen letter pair, *Dirichlet smoothing* (add-α pseudo-counts) is exactly this table's second row.

Checkpoint: the one-line update

Q QUESTION

Prior Beta$(2, 3)$; you then observe 4 heads and 1 tail. The posterior is...

- **A.** Beta$(6, 4)$
- **B.** Beta$(3, 7)$
- **C.** Beta$(5, 2)$
- **D.** Beta$(7, 8)$

Answer: the one-line update

A ANSWER

Correct: A — Beta$(2+4, 3+1) =$ **Beta**$(6,4)$

Why: Heads land on α, tails on β — nothing else moves. B swapped the two counts; C threw the prior away and updated a flat Beta$(1,1)$; D added the total $n=5$ to both dials, double-counting every flip.

Watching beliefs sharpen

Today's posterior is tomorrow's prior

Nothing forces the flips to arrive in one batch. Feed them one at a time, and **each posterior becomes the next prior**:

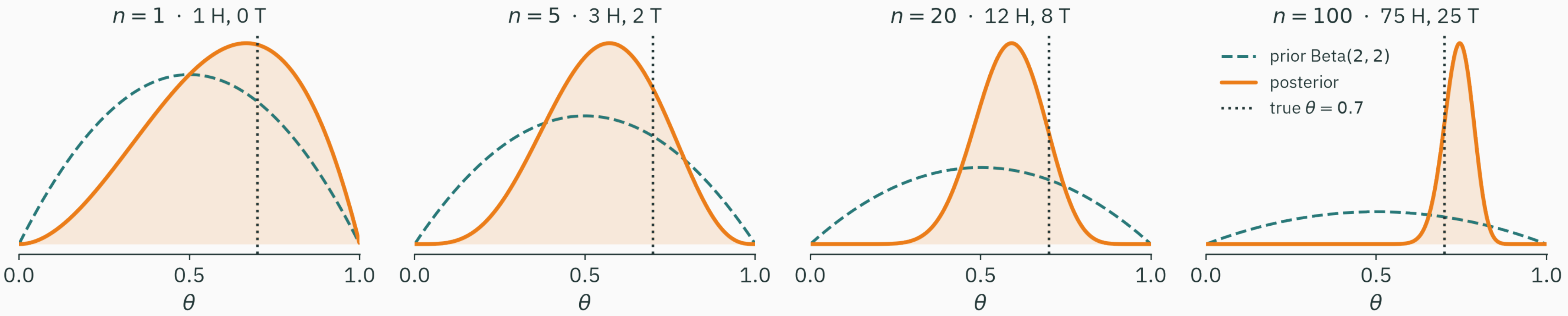
$$\mathrm{Beta}(\alpha,\beta) \xrightarrow{\mathrm{H}} \mathrm{Beta}(\alpha+1,\beta) \xrightarrow{\mathrm{T}} \mathrm{Beta}(\alpha+1,\beta+1) \xrightarrow{\mathrm{H}} \ldots$$

```python
a, b = 2, 2                         # Beta(2,2): 1 imaginary head, 1 tail
for y in flips:                     # y = 1 for heads, 0 for tails
    a, b = a + y, b + 1 - y         # the ENTIRE Bayesian update
# at any moment: posterior = Beta(a, b), MAP = (a-1)/(a+b-2)
```

Two integers of memory, one addition per observation. This is **online learning** in its purest form — the model never re-reads old data.

One hundred flips, four snapshots

A coin with true $\theta = 0.7$ streams flips into a Beta$(2, 2)$ prior:



After 1 flip the prior still dominates; by 100 flips (75 H, 25 T → Beta$(77, 27)$) the posterior is a needle — parked at the **sample's** rate 0.75, a whisker above the truth. That residual wobble is the data's own sampling noise, not the prior's doing.

The posterior sharpens like $1/\sqrt{n}$

How fast is certainty earned? The posterior's spread obeys

$$\mathrm{sd} \approx \sqrt{\frac{\hat{\theta}\left(1-\hat{\theta}\right)}{n}}$$

- Quadruple the flips → **halve** the uncertainty — the same $\sqrt{n}$ law you met in L13's CLT demo, now running *inside* the posterior.
- By $n = 100$, the prior's 4 pseudo-flips are 4 votes in an electorate of 104: present, out-voted, and fading.

Order cannot matter

Flip H then T, starting from Beta(α, β): $(\alpha + 1, \beta)$, then $(\alpha + 1, \beta + 1)$. Flip T then H: $(\alpha, \beta + 1)$, then $(\alpha + 1, \beta + 1)$. **Same place.**

No accident: the posterior is a **product** of per-flip factors, and multiplication commutes —

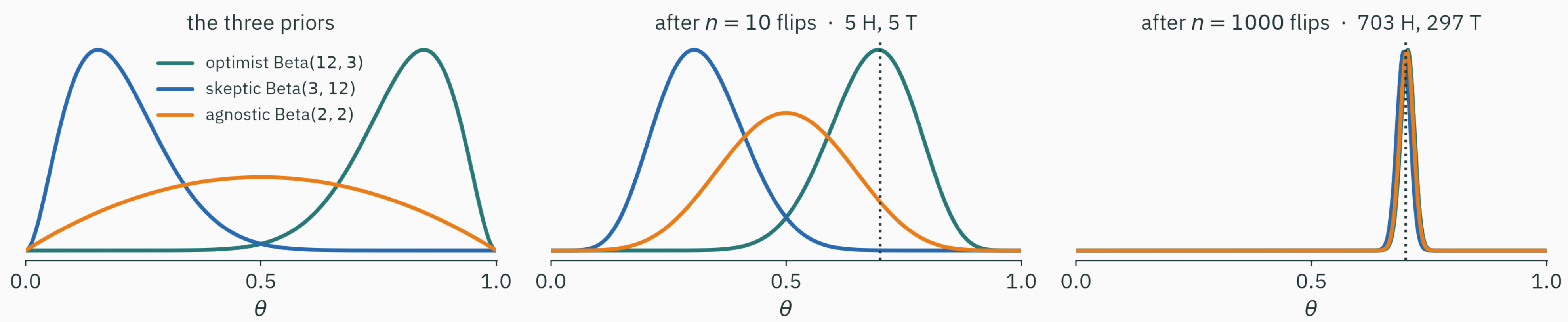
$$P(\theta \mid D) \propto P(\theta) \prod_{i=1}^{n} P(y_i \mid \theta)$$

Stream the data or dump it in one batch, in any order — exactly the same posterior.

Three scientists, three priors

V VISUAL

An optimist, a skeptic, and an agnostic watch the same coin ($\theta = 0.7$):



Ten flips in (5 H, 5 T): they still disagree — the ambiguous data lets each prior shine through. A thousand flips in: three posteriors, one needle. **They argued; the coin adjudicated.**

When the prior washes out

The formula says exactly when belief stops mattering:

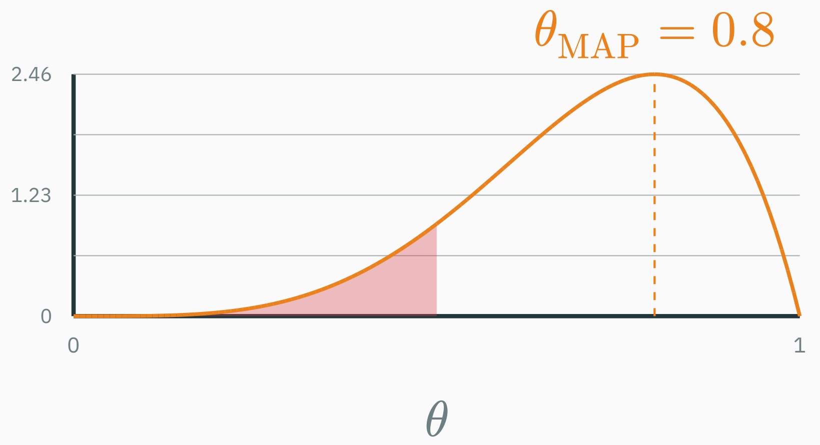
$$\theta_{\text{MAP}} = \frac{n_H + \alpha - 1}{n + \alpha + \beta - 2} \to \frac{n_H}{n} = \theta_{\text{MLE}} \quad \text{as } n \to \infty$$

- The prior is a **fixed** number of pseudo-votes in a growing electorate: decisive at $n = 3$, a rounding error at $n = 10^6$.
- Scarce data → the prior carries you. Abundant data → MLE and MAP agree and the argument dissolves.

Priors are training wheels that remove themselves — at a $1/n$ rate you can compute.

Posterior inference for the opening coin example

Three heads in three flips, Beta$(2, 2)$ prior → posterior Beta$(5, 2)$:



- The MAP estimate is $\theta_{\text{MAP}} = 0.8$, while the posterior still assigns about 11% probability to $\theta < 0.5$.
- Three flips buy you suspicion, never certainty — and never, ever “tails is impossible”. The absurd bet is off.

Slide the prior yourself

INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/mle-map-coin

A coin you control: set the true bias, choose α, β, and flip. Watch θ_{MAP} glide from the prior's peak toward θ_{MLE} as the pseudo-counts get out-voted.

Try the failure modes: a strong wrong prior (how much data undoes it?), and $n = 3$ with a flat prior (the absurd bet, live).

Gaussian meets Gaussian

Same movie, continuous star

New estimation problem: a sensor reads a true value μ through Gaussian noise (L13):

$$x_i \sim \mathcal{N}(\mu, \sigma^2), \quad \sigma \text{ known}, \quad i = 1, \dots, n$$

You also have a hunch — the calibration sheet says readings sit near μ_0:

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

Beta was the prior for a **probability**; the Gaussian is the prior for a **location**. And the Gaussian is conjugate to itself: Gaussian in → Gaussian out.

Stated: Gaussian in, Gaussian out

Multiply the two bells and complete the square (algebra in T7 and MML 8.3.2 — today, the shape of the answer). Call $1/\text{variance}$ **precision** — confidence, as a number: prior precision $\tau_0 = 1/\sigma_0^2$, data precision $\tau_D = n/\sigma^2$.

$$P(\mu \mid D) = \mathcal{N}(\mu_n, \sigma_n^2), \qquad \frac{1}{\sigma_n^2} = \tau_0 + \tau_D$$

$$\mu_n = \frac{\tau_0\,\mu_0 + \tau_D\,\bar{x}}{\tau_0 + \tau_D}$$

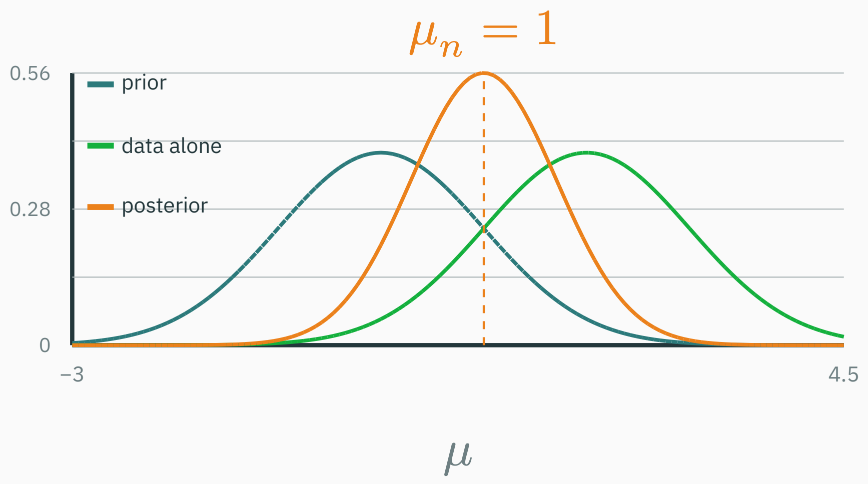
Precisions add, and the posterior mean is a **precision-weighted average** of hunch μ_0 and data average $\bar{x}$ — the surer voice counts more.

Picture and numbers

V VISUAL

Hunch $\mu_0 = 0$ (precision $1/\sigma_0^2 = 1$); data $n = 4$, $\bar{x} = 2$, $\sigma^2 = 4$ (precision $n/\sigma^2 = 1$). Equal confidence → meet exactly halfway:

$$\mu_n = \frac{1 \cdot 0 + 1 \cdot 2}{1 + 1} = 1, \qquad \sigma_n^2 = \frac{1}{1+1} = 0.5$$



The posterior is **narrower than both parents** — two independent confidences stacked.

Precision is the currency

- Posterior mean $=$ average of the two centers, **weighted by precision** — the surer voice counts more.
- The prior behaves like σ^2/σ_0^2 **virtual observations** pinned at μ_0 — pseudo-counts again, in continuous clothing.
- As n grows, data precision $n/\sigma^2 \to \infty$: the hunch washes out — the same story the coin just told.

Conjugate updating is an average where precision is the currency: pseudo-flips for coins, virtual readings for means.

Every regularizer is a prior

L14′s keystone, one more time

Linear regression, Gaussian noise: $y_i = \boldsymbol{w}^\top \boldsymbol{x}_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. L14 showed:

$$\mathrm{NLL}(\boldsymbol{w}) = \frac{1}{2\sigma^2}\|\boldsymbol{y} - X\boldsymbol{w}\|^2 + \mathrm{const}$$

Minimizing NLL **is** least squares — **MSE = Gaussian NLL**. That was keystone 1 of this module.

So far, pure MLE: the weights $\boldsymbol{w}$ enjoy a flat prior nobody admitted to. Time to admit a real one.

Give the weights a prior

Belief: weights shouldn't be huge — a model that needs $w = 10^6$ is probably memorizing noise. Encode it:

$$\boldsymbol{w} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 I) \quad \Rightarrow \quad -\log P(\boldsymbol{w}) = \frac{1}{2\sigma_0^2}\|\boldsymbol{w}\|^2 + \text{const}$$

MAP swaps argmax-posterior for argmin of the negative log-posterior:

$$\boldsymbol{w}_{\text{MAP}} = \arg\min_{\boldsymbol{w}} \underbrace{\frac{1}{2\sigma^2}\|\boldsymbol{y} - X\boldsymbol{w}\|^2}_{\text{data misfit (L14)}} + \underbrace{\frac{1}{2\sigma_0^2}\|\boldsymbol{w}\|^2}_{\text{prior's fine on big weights}}$$

```
nll  = ((y - X @ w)**2).sum() / (2 * sig2)      # L14: Gaussian NLL
nlp  = (w**2).sum() / (2 * sig0_2)              # -log N(0, sig0^2) prior
loss = nll + nlp                                # MAP objective
```

You have seen this loss before — or you will

Multiply through by $2\sigma^2$ (argmins don't care):

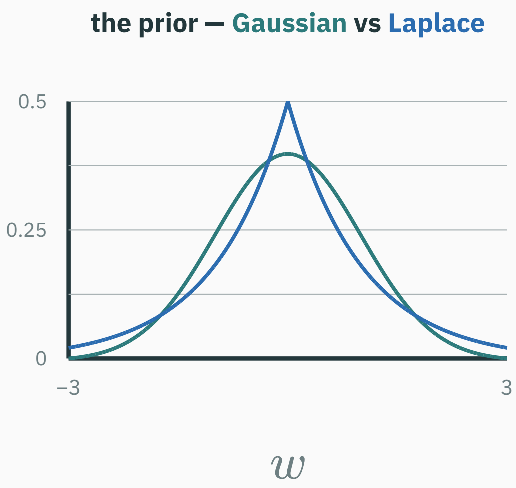
$$\boldsymbol{w}_{\mathrm{MAP}} = \arg\min_{\boldsymbol{w}} \|\boldsymbol{y} - X\boldsymbol{w}\|^2 + \lambda\|\boldsymbol{w}\|^2, \qquad \lambda = \frac{\sigma^2}{\sigma_0^2}$$

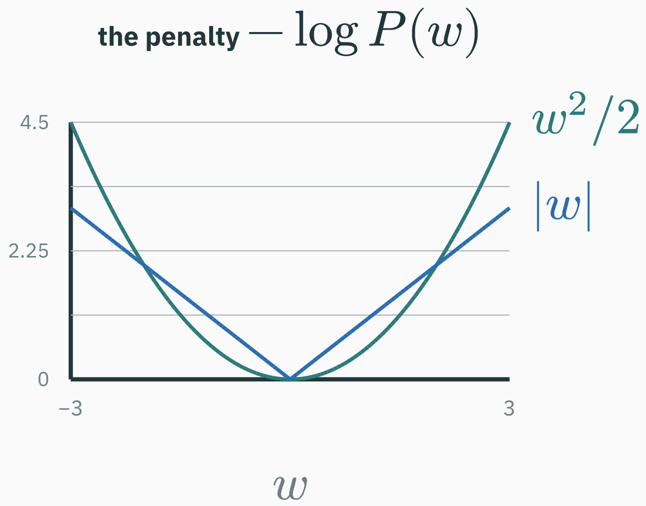
Ridge regression is MAP with a Gaussian prior on the weights. The penalty knob λ is the noise-to-prior-confidence ratio — keystone 2

Promissory note: your ML course will hand you "ridge" and "weight decay" as recipes, and tune λ by validation. You now hold the receipt: cranking λ **is** shrinking σ_0 — declaring harder that big weights are unbelievable.

Bell to bowl

$-\log$ turns a belief into a penalty — every prior shape mints a regularizer:





Gaussian prior → quadratic penalty → **L2 / ridge**. Laplace prior → $|w|$ penalty → **L1 / lasso**; the corner at zero allows exact zero-valued coefficients.

The regularization dictionary

You add to the loss...	You just assumed...	Its street name
$\lambda\lVert\boldsymbol{w}\rVert^2$	Gaussian prior $\mathcal{N}(0, \sigma_0^2)$	ridge / weight decay
$\lambda\lVert\boldsymbol{w}\rVert_1$	Laplace prior	lasso
add-α pseudo-counts	Beta / Dirichlet prior	Laplace smoothing (→ L26)

Choosing a regularizer **is** choosing a prior. There is no prior-free learning — only unexamined priors.

What to take home

Lecture 15 — summary

- **Bayes**: posterior $\propto$ likelihood $\times$ prior — the evidence only normalizes.
- **MAP** climbs log-likelihood $+$ log-prior; flat prior $\Rightarrow$ MAP $=$ MLE.
- **Beta(α, β)**: $\alpha - 1$ imaginary heads, $\beta - 1$ imaginary tails; $\alpha + \beta$ is the stake.
- **Conjugacy** ⭐: Beta prior $+$ coin data $\rightarrow$ Beta$(\alpha + n_H, \beta + n_T)$ — two additions.
- **Sequential $=$ batch**, any order; width $\sim 1/\sqrt{n}$; data swamps the prior.
- **Gaussians**: precisions add; posterior mean $=$ precision-weighted average.
- **Regularizers are priors**: Gaussian → ridge, Laplace → lasso, counts → smoothing.

Read before L16 — MML 6.6.1 & 8.3.2 · MacKay Ch. 3. **T7** pairs with this lecture (grid-check every derivation); **Quiz 3 / midsem** covers Module 3.

Practice problems

Try on paper; verify with the T7 notebook's grid trick.

1. Re-derive the ⭐ update: Beta(α, β) prior, n_H heads, n_T tails → posterior, from a blank page.
2. Prior Beta$(5, 5)$, data 2 H, 8 T: posterior, θ_{MAP}, posterior mean — and compare with θ_{MLE}.
3. Show that Beta$(1, 1)$ makes $\theta_{\text{MAP}} = \theta_{\text{MLE}}$ for **every** possible dataset.
4. How many **consecutive heads** push θ_{MAP} above 0.99 under a Beta$(2, 2)$ prior? (Solve $(n + 1)/(n + 2) > 0.99$.)
5. Sensor: prior $\mathcal{N}(0, 1)$; data $n = 9$, $\bar{x} = 3$, $\sigma^2 = 9$. Posterior mean and variance?
6. A colleague doubles ridge's λ. What just happened to σ_0^2 in the prior story — and which prior would have produced an L1 penalty instead?

⭐⭐⭐ MAP keeps the peak, throws the width

★ OPTIONAL

MAP reports **one number** — but the posterior was a whole curve, and the curve knows more.

- Predicting the next flip by **averaging over the posterior** gives the posterior *mean*, not the mode: $P(\text{next H} \mid D) = \frac{\alpha+n_H}{\alpha+\beta+n}$.
- Three heads, Beta$(2,2)$: predict $5/7 \approx 0.71$ — milder than $\theta_{\text{MAP}} = 0.8$, and a world away from MLE's 1.0.
- Keeping the **entire** posterior (and integrating over it) is Bayesian machine learning proper — the Probabilistic ML course lives there.

⭐⭐⭐ Laplace's rule of succession

★ OPTIONAL

Flat prior + posterior mean → $P(\text{next H}) = \frac{n_H+1}{n+2}$ — "add one imaginary head and one imaginary tail."

- Laplace, 1774: what is the probability the sun rises tomorrow, given n recorded sunrises? Answer: $(n+1)/(n+2)$ — huge, but **never 1**.
- NLP's **add-one smoothing** is this rule, verbatim, applied to word counts — and its add-α generalization is the Dirichlet machinery L26 uses to keep the language model's $\log 0$ at bay.

250 years old, still shipping in production systems. Priors age well.

Every regularizer is a prior in disguise.

Next: **Module 4 — The Optimization Landscape**. We now know *what* to minimize; L16 begins the story of *how*.